Vibrational constants of AlO from its electronic band spectrum

General Physics Laboratory V, experiment 8 - Department of Physics and Electronics, CHRIST (Deemed to be University), Bengaluru

Name	**A. Student**
Register number	**2447101**
Batch	**-**
Date	**2026-09-12**
Least count of the comparator	**0.001 cm**

Aim

To determine the vibrational constants of the aluminium oxide molecule for the upper ($B^2\Sigma^+$) and lower ($X^2\Sigma^+$) electronic states from the band heads of its visible band system.

Observations

The mercury comparison spectrum, with the lines measured marked and labelled by wavelength in Å.

Table 1 - comparator readings of the mercury standard lines

Line	λ (Å)	M.S.R. (cm)	V.S.D.	d (cm)
Hg Violet	4046.56	2.85	1	2.851
Hg Violet (weak)	4077.83	3.35	0	3.350
Hg Blue	4358.33	7.00	47	7.047
Hg Blue-green (weak)	4916.04	11.75	40	11.790
Hg Green	5460.74	14.65	15	14.665
Hg Yellow I	5769.60	15.85	13	15.863
Hg Yellow II	5790.66	15.90	36	15.936

$$\lambda = \lambda_0 + C / (d - d_0)$$

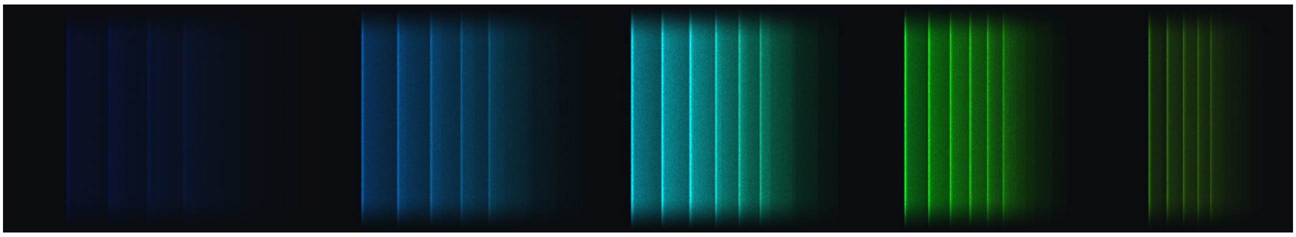
Constants worked out from three mercury lines: λ_0 = 2554.35 Å, C = -36231.3 Å·cm, d_0 = 27.1312 cm

Working: R = (l1-l2)(d3-d2) / (l2-l3)(d2-d1) = 1.8423, d0 = (R d1 - d3)/(R - 1) = 27.1298 cm, then C from the first pair and l0 from the first line.

Table 2 - the same constants applied to every mercury line measured

Line	d (cm)	λ calculated (Å)	λ standard (Å)	difference (Å)
Hg Violet	2.851	4046.57	4046.56	0.01
Hg Violet (weak)	3.350	4077.88	4077.83	0.05
Hg Blue	7.047	4358.32	4358.33	-0.01

Line	d (cm)	λ calculated (Å)	λ standard (Å)	difference (Å)
Hg Blue-green (weak)	11.790	4916.05	4916.04	0.01
Hg Green	14.665	5460.71	5460.74	-0.03
Hg Yellow I	15.863	5769.71	5769.60	0.11
Hg Yellow II	15.936	5790.67	5790.66	0.01



The AlO band system as recorded on the plate. Each mark is a band head measured with the comparator.

Table 3 - AlO band heads

Band (v', v")	Δv	d (cm)	λ (Å)	ν̃ (cm^{-1})
(2, 0)	+2	8.217	4469.90	22371.8
(3, 1)	+2	8.446	4493.38	22255.0
(4, 2)	+2	8.657	4515.53	22145.8
(5, 3)	+2	8.853	4536.56	22043.2
(1, 0)	+1	9.824	4647.77	21515.7
(2, 1)	+1	10.018	4671.50	21406.4
(3, 2)	+1	10.198	4694.00	21303.8
(4, 3)	+1	10.363	4715.06	21208.7
(5, 4)	+1	10.516	4734.95	21119.5
(0, 0)	0	11.293	4841.93	20652.9
(1, 1)	0	11.459	4866.16	20550.1
(2, 2)	0	11.611	4888.80	20454.9
(3, 3)	0	11.751	4910.05	20366.4
(4, 4)	0	11.878	4929.67	20285.4
(5, 5)	0	11.994	4947.87	20210.7
(0, 1)	-1	12.783	5079.49	19687.0
(1, 2)	-1	12.912	5102.39	19598.6
(2, 3)	-1	13.030	5123.72	19517.1
(3, 4)	-1	13.136	5143.18	19443.2
(4, 5)	-1	13.232	5161.06	19375.9
(5, 6)	-1	13.318	5177.29	19315.1
(0, 2)	-2	14.112	5337.25	18736.2
(1, 3)	-2	14.211	5358.58	18661.7
(2, 4)	-2	14.300	5378.03	18594.2

Band (v', v")	Δv	d (cm)	λ (Å)	ṽ (cm^{-1})
(3, 5)	-2	14.379	5395.52	18533.9
(4, 6)	-2	14.448	5410.98	18481.0

Deslandre tables

Table 4 - band heads in Å

v' \ v"	0	1	2	3	4	5	6
0	4841.9	5079.5	5337.3	-	-	-	-
1	4647.8	4866.2	5102.4	5358.6	-	-	-
2	4469.9	4671.5	4888.8	5123.7	5378.0	-	-
3	-	4493.4	4694.0	4910.1	5143.2	5395.5	-
4	-	-	4515.5	4715.1	4929.7	5161.1	5411.0
5	-	-	-	4536.6	4734.9	4947.9	5177.3

Table 5 - band heads in cm^{-1}

v' \ v"	0	1	2	3	4	5	6
0	20652.9	19687.0	18736.2	-	-	-	-
1	21515.7	20550.1	19598.6	18661.7	-	-	-
2	22371.8	21406.4	20454.9	19517.1	18594.2	-	-
3	-	22255.0	21303.8	20366.4	19443.2	18533.9	-
4	-	-	22145.8	21208.7	20285.4	19375.9	18481.0
5	-	-	-	22043.2	21119.5	20210.7	19315.1

Upper electronic state $B^2\Sigma^+$

First differences ΔG'(v + ½), cm^{1}

Interval	Value taken
ΔG'(0 + ½)	862.8
ΔG'(1 + ½)	856.0
ΔG'(2 + ½)	848.9

Second difference and the constants

Quantity	Value
$\Delta^2 G' = \Delta G'(½) - \Delta G'(3/2)$	6.80
ωe'xe' (cm^{-1})	3.40
ωe' (cm^{-1})	869.6
xe'	0.00391

Lower electronic state $X^2\Sigma^+$

First differences ΔG"(v + ½), cm^{1}

Interval	Value taken
ΔG"(0 + ½)	965.6
ΔG"(1 + ½)	951.3
ΔG"(2 + ½)	937.3

Second difference and the constants

Quantity	Value
Δ^2G" = ΔG"(½) -ΔG"(3/2)	14.30
ωe"xe" (cm^{-1})	7.15
ωe" (cm^{-1})	979.9
xe"	0.00730

Result

Table 6 - vibrational constants of AlO

Quantity	My value	Literature	Difference (%)
ωe' (cm^{-1}), upper state $B^2\Sigma^+$	869.6	870.0	0.1
ωe'xe' (cm^{-1})	3.40	3.52	3.4
xe'	0.00391	0.00405	3.4
ωe" (cm^{-1}), lower state $X^2\Sigma^+$	979.9	979.2	0.1
ωe"xe" (cm^{-1})	7.15	6.97	2.6
xe"	0.00730	0.00712	2.6

The vibrational quantum of the excited state is smaller and its anharmonicity larger, which is what one expects: the bond is weaker and the potential well shallower once the molecule is electronically excited.

Sources of error

Setting the crosswire on a shaded head is the largest single source of error; the two yellow mercury lines are close enough that the field had to be narrowed to separate them.

Questions

1. What do you mean by an electronic band spectrum?

Answer 1 as written in the record book.

2. What do you mean by the vibrational constants of AlO?

Answer 2 as written in the record book.

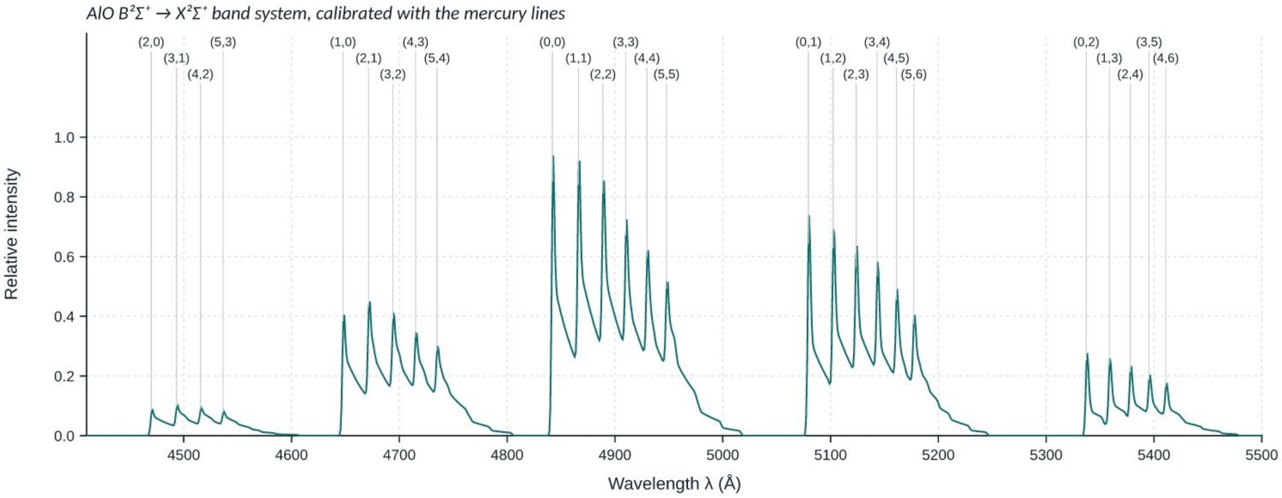
3. Do you expect a band spectrum from a homonuclear diatomic molecule?

Answer 3 as written in the record book.

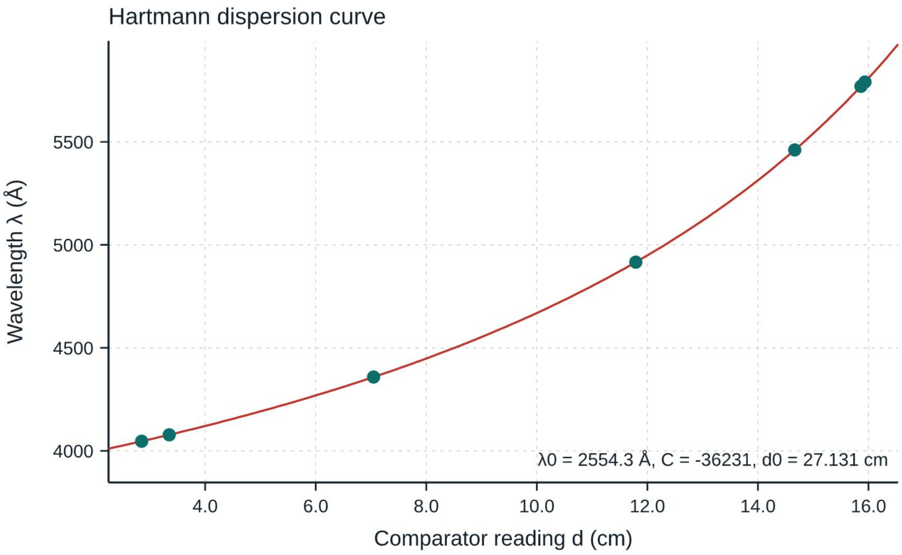
4. Explain sequences and progressions.

Answer 4 as written in the record book.

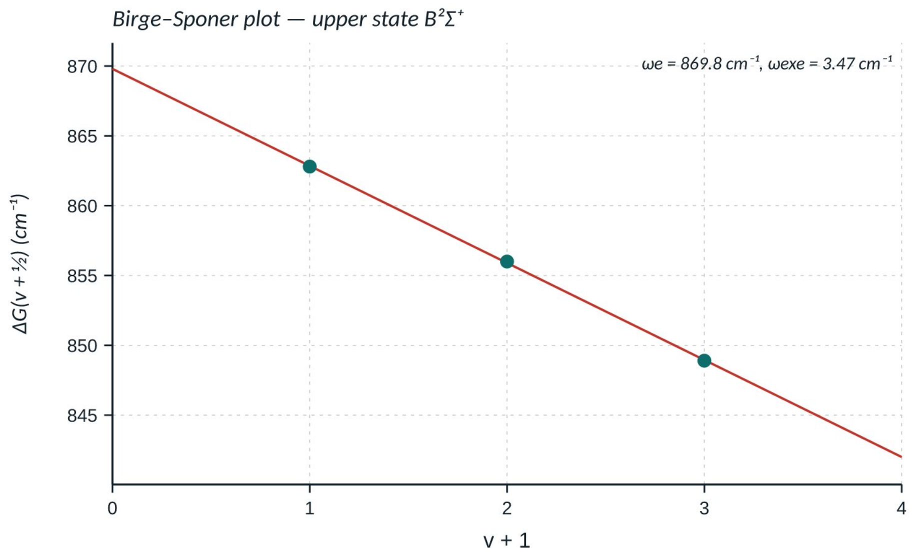
Graphs



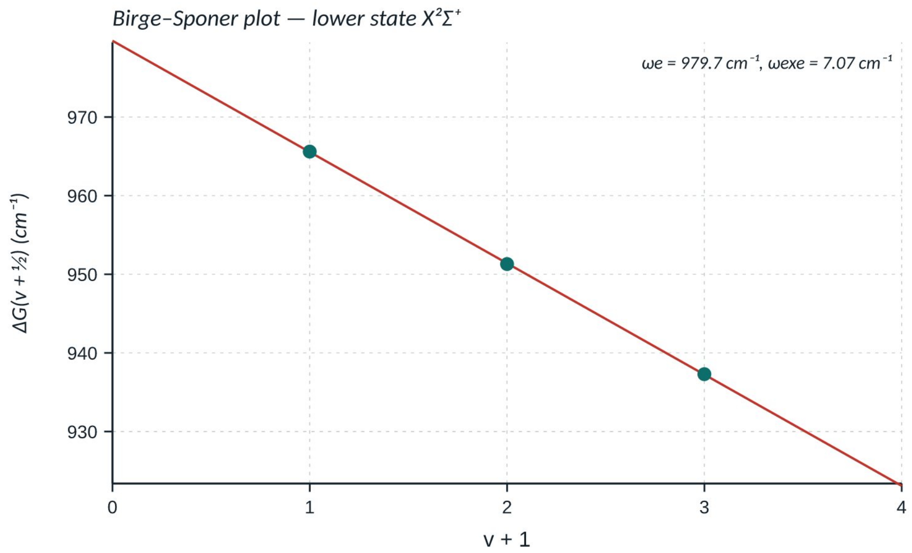
The densitometer trace of the same plate, on a wavelength scale from your own calibration.



Hartmann dispersion curve through the mercury lines.



Birge-Sponer plot for the upper state.



Birge-Sponer plot for the lower state.